

What is Generative AI

What is Generative AI and How Transformers Make it Possible

An Introduction to Generative AI and the Role of Transformers

By Bhaskar jha | Feb 29, 2024

Recent breakthroughs in the field of GEN AI are taking the world by storm and have the potential to change how we approach content creation drastically. Generative AI is a machine's ability to create new content, including text, images, audio, video, code, and simulations.

Generative AI systems fall under the broad category of Artificial Intelligence and Machine learning.

What is GEN AI?

Gen AI is a subset of Deep learning (a subset of Machine learning). It uses artificial neural networks and can process labeled and unlabeled data using supervised, unsupervised, and semi-supervised methods.

Machine learning models, in general, can be divided into two types:

1. Discriminative Models: These models focus on the relationship between features and labels. For instance, if we have data on animal features and their names, a discriminative model would learn the boundary between different animals. When given a new set of features, it would predict which animal they belong to.
2. Generative Models: These models learn the underlying distribution of the data. They don't just learn the boundary; they learn how the data itself is generated. This allows them to create new, synthetic data that follows the same patterns as the training data. For example, a generative model trained on images of cats could generate entirely new, realistic images of cats.

How Transformer makes it Possible: An introduction

Before Transformers, traditional NLP models like Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) processed text sequentially. They would read one word at a time, keeping track of previous words to understand the context. However, this sequential processing had several limitations:

1. Difficulty in capturing long-range dependencies: As sentences got longer, it became harder for RNNs to remember information from the beginning of the sentence.
2. Computational inefficiency: Sequential processing is slow because each word depends on the previous one, making it difficult to parallelize training.

How does the Transformer work?

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Introduced in the 2017 paper "Attention Is All You Need," the Transformer architecture revolutionized NLP by moving away from sequential processing and introducing the "Attention Mechanism."

Key components include:

- * Self-Attention: Allows the model to look at other words in the same sentence to understand the context of a particular word. For example, in the sentence "The animal didn't cross the street because it was too tired," the mechanism helps the model understand that "it" refers to "the animal."
- * Parallelization: Unlike RNNs, Transformers can process all words in a sentence simultaneously, making training much more efficient on large datasets.
- * Positional Encoding: Since Transformers don't process text sequentially, they use positional encoding to understand the order of words in a sentence.

At a high level, a transformer model consists of an encoder and a decoder. The encoder converts input text into an intermediate representation, and the decoder converts that representation into useful text. Unlike previous models, the Transformer understands context through its self-attention mechanism, weighing the importance of different positions in the input sequence.

Hallucinations

Hallucinations are words or phrases generated by AI that are often nonsensical or grammatically incorrect. These occur due to:

1. Insufficient training data.
2. Training on noisy or "dirty" data.
3. Lack of context or constraints.

Hallucinations can be problematic because they make the output difficult to understand and can lead the model to generate incorrect or misleading information.